Exercises

Exercise 3.1 · Exponential forms. Write in exponential form: $-2+2i$; $-1-i\sqrt{3}$; $5i$; $\dfrac{1+i}{1-i}$.

Exercise 3.2 · Two ways of writing the same product. Let $z_1=\sqrt{3}+i$ and $z_2=1-i$.

1. Write z_1, z_2 and then z_1z_2 in exponential form.
2. Write z_1z_2 in algebraic form.
3. Deduce the exact values of $\cos\dfrac{\pi}{12}$ and $\sin\dfrac{\pi}{12}$.

Exercise 3.3 · Powers.

1. Compute $(1-i)^8$ and $\left(\sqrt{3}+i\right)^{12}$.
2. Determine the natural numbers $n\geq 1$ for which $\left(\sqrt{3}+i\right)^n$ is real; positive real; purely imaginary. Give the smallest in each case.

Exercise 3.4 · De Moivre's formula. Using de Moivre's formula and the binomial formula, express $\cos 4\theta$ in terms of $\cos\theta$ only, and $\sin 4\theta$ in terms of $\cos\theta$ and $\sin\theta$.

Exercise 3.5 · Linearising in order to integrate.

1. Using Euler's formulas, prove that $\sin^3 x=\dfrac{3\sin x-\sin 3x}{4}$.
2. Deduce the value of $\displaystyle\int_0^{\pi}\sin^3 x\,\mathrm{d}x$.

Exercise 3.6 · Modulus and argument of $1+\mathrm{e}^{i\theta}$.

1. Prove that for every real number θ, $1+\mathrm{e}^{i\theta}=2\cos\dfrac{\theta}{2}\,\mathrm{e}^{i\theta/2}$.
2. Deduce the modulus and an argument of $1+\mathrm{e}^{i\theta}$ when $\theta\in(-\pi,\pi)$.
3. Application: modulus and argument of $1+\mathrm{e}^{2i\pi/3}$, then a check with the algebraic form.

Exercise 3.7 · A sum of cosines. Let θ be a real number that is not a multiple of 2π, and n a natural number.

1. Prove that $\displaystyle\sum_{k=0}^{n}\mathrm{e}^{ik\theta}=\frac{\mathrm{e}^{i(n+1)\theta}-1}{\mathrm{e}^{i\theta}-1}$.
2. Using the half-angle method, prove that this sum equals $\mathrm{e}^{in\theta/2}\dfrac{\sin\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}$.

3. Deduce $\sum_{k=0}^{n} \cos(k\theta) = \dfrac{\cos\frac{n\theta}{2}\ \sin\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}$, then check the formula for $n = 2$ and $\theta = \frac{\pi}{2}$.